

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Kinematics Graphs

Graph questions ask for gradients, areas, and clean piecewise stories.

KINEMATICS GRAPHS · M1 · 03

ELITEIGCSE.COM · WHATSAPP +20 112 000 9622



DR ESLAM AHMED · CAIRO UNIVERSITY FACULTY OF ENGINEERING

The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 GRAD: Gradient Means Acceleration

TRIGGER *velocity-time graph gradient*

02 AREA: Area Means Displacement

TRIGGER *area under velocity-time graph*

03 PIECE: Piecewise Motion

TRIGGER *multi-stage graph*

04 SKETCH: Sketch From Words

TRIGGER *draw velocity-time graph*

BANK EVIDENCE Local pages: 59 problem, 86 answer, 17 notes. Website primary entries: 12. Website linked entries: 17.

01 GRAD: Gradient Means Acceleration

TRIGGER *velocity-time graph gradient*

BECOMES Acceleration is the gradient of the velocity-time graph.

FIRST LINE TO WRITE

$$a = \frac{\Delta v}{\Delta t}$$

SIMPLEST STRATEGY

1. Get two points.
2. Rise is velocity change.
3. Across is time change.
4. Divide with sign.

WORKED MODEL

▮ A fall from 12 to 4 in 2 s gives $a = -4 \text{ m s}^{-2}$.

HARD VARIANTS

1. Curved graphs need tangent gradients.
2. Use signs: below the time axis means negative velocity.
3. Acceleration changes on each different line segment.

— BOTTOM LINE

Downward graph slope means negative acceleration.

COMMON TRAP Using distance over time from a velocity-time graph.

02 AREA: Area Means Displacement

TRIGGER *area under velocity-time graph*

BECOMES Displacement is signed area under the velocity-time graph.

FIRST LINE TO WRITE

$$s = \text{area under } v-t$$

SIMPLEST STRATEGY

1. Add simple shapes.
2. Record sign above or below axis.
3. Evaluate each area.
4. Aggregate displacement.

WORKED MODEL

▮ A triangle of base 6, height 10 has area 30.

HARD VARIANTS

1. For total distance, add absolute areas.
2. If the graph crosses the axis, split the area at the crossing.
3. Use trapeziums for sloping segments with two non-zero velocities.

— BOTTOM LINE

Area gives displacement, not always distance.

COMMON TRAP Calling signed displacement total distance.

03 PIECE: Piecewise Motion

TRIGGER *multi-stage graph*

BECOMES Treat each graph segment as its own motion stage.

FIRST LINE TO WRITE

| $\text{stage 1} + \text{stage 2} + \dots$

SIMPLEST STRATEGY

1. Partition at corners.
2. Identify each shape.
3. Evaluate gradients/areas.
4. Combine carefully.
5. Explain the motion in words.

WORKED MODEL

| Constant velocity is a horizontal line; constant acceleration is a straight sloping line.

HARD VARIANTS

1. Rest is $v = 0$, not necessarily $s = 0$.
2. Negative velocity means motion in the opposite direction.
3. Average speed needs total distance divided by total time.

— BOTTOM LINE

Every corner starts a new rule.

COMMON TRAP Using one acceleration for the whole graph.

04 SKETCH: Sketch From Words

TRIGGER *draw velocity-time graph*

BECOMES Convert each phrase into a graph feature.

FIRST LINE TO WRITE

| accelerates uniformly $\Rightarrow$ straight sloping line

SIMPLEST STRATEGY

1. Start at the given velocity.
2. Keep axes labelled.
3. Encode each phrase.
4. Time-scale the segments.
5. Check areas if distance is given.
6. Hold units.

WORKED MODEL

| Uniform deceleration to rest is a straight line ending on the time axis.

HARD VARIANTS

1. If distance is given, area must match it.
2. If acceleration is greater, slope must be steeper.
3. Graph labels can earn marks even when scale is not perfect.

— BOTTOM LINE

The shape tells the motion.

Sketch a v - t graph for acceleration, constant speed, then rest.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
velocity-time graph gradient	$a = \frac{\Delta v}{\Delta t}$
area under velocity-time graph	$s = \text{area under } v-t$
multi-stage graph	stage 1 + stage 2 + ...
draw velocity-time graph	accelerates uniformly $\Rightarrow$ straight sloping line

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
GRAD: Gradient Means Acceleration	Curved graphs need tangent gradients. Use signs: below the time axis means negative velocity. Acceleration changes on each different line segment.
AREA: Area Means Displacement	For total distance, add absolute areas. If the graph crosses the axis, split the area at the crossing. Use trapeziums for sloping segments with two non-zero velocities.
PIECE: Piecewise Motion	Rest is $v = 0$, not necessarily $s = 0$. Negative velocity means motion in the opposite direction. Average speed needs total distance divided by total time.
SKETCH: Sketch From Words	If distance is given, area must match it. If acceleration is greater, slope must be steeper. Graph labels can earn marks even when scale is not perfect.